



fixtSE DIY Smart Sensor



VIEW IN BROWSER

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Summary

This case is designed to house an HLK-LD2450 mmWave Radar Sensor and an ESP32 30-pin board. No soldering skills needed

[Hobby & Makers](#) > [Electronics](#)

This remix of the [esp32-wroom case](#) and the [everything presence lite](#) it's a versatile, pressure-fit wonder, requiring zero soldering or specialized tools; just snap that ESP32 Board together with the HLK-LD2450 sensor. Then, flash it with ESPHome using our custom firmware, and voilà – you've created the foundation for your smart sensor!

With enough space left in the case, you can effortlessly expand its capabilities by incorporating a microphone, light sensor, temperature sensor, air quality sensor, or any other sensor that you want.

Printed with PLA, 40% infill.

Watch the full tutorial here: <https://fixtse.com/blog/esphome-mmwave-sensor>

You can buy the devices used in the project here:

Using the affiliate links below helps support all the content I create.

- ESP32 30 PIN Development Board
 - Amazon: ([3 pack](#))

- Aliexpress: ([USB C / Micro USB](#))
- HLK-LD2450 mmWave Radar Sensor
 - Amazon: ([HLK-LD2450 Kit](#))
 - Aliexpress: (HLK-LD2450 Kit)
 - [Option 1](#)
 - [Option 2](#)

Optional

- Dupont Cable
 - Amazon: ([Dupont Cable Female to female](#))
 - Aliexpress: ([Dupont Cable Female to female](#))
- USB C Cable
 - Amazon: ([2 Pack](#))
 - Aliexpress: ([USB C Cable](#))
- 5v charger
 - Amazon: ([2 Pack](#))
 - Aliexpress: ([5v charger](#))

If you plan on adding a light sensor in the future, print the case in white (black reduces the light captured by the sensor, so it will report lower values)

This remix is based on



Everything Presence Lite - Official Case

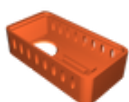
by Everything Smart Home



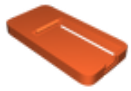
esp32-wroom case - Temp and mmWave - DHT11 LD2410

by djc3d

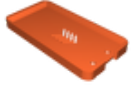
Model files



body.stl



esp32_housing.stl



esp32_lid.stl



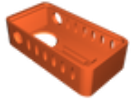
joint.stl



base.stl



locknut.stl



body_light_temperature.stl

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